AI for Energy:
Data-efficient and explainable algorithms to exploit energy flexibility

Chris Develder, et al.
AI4E Team, IDLab, Ghent University – imec

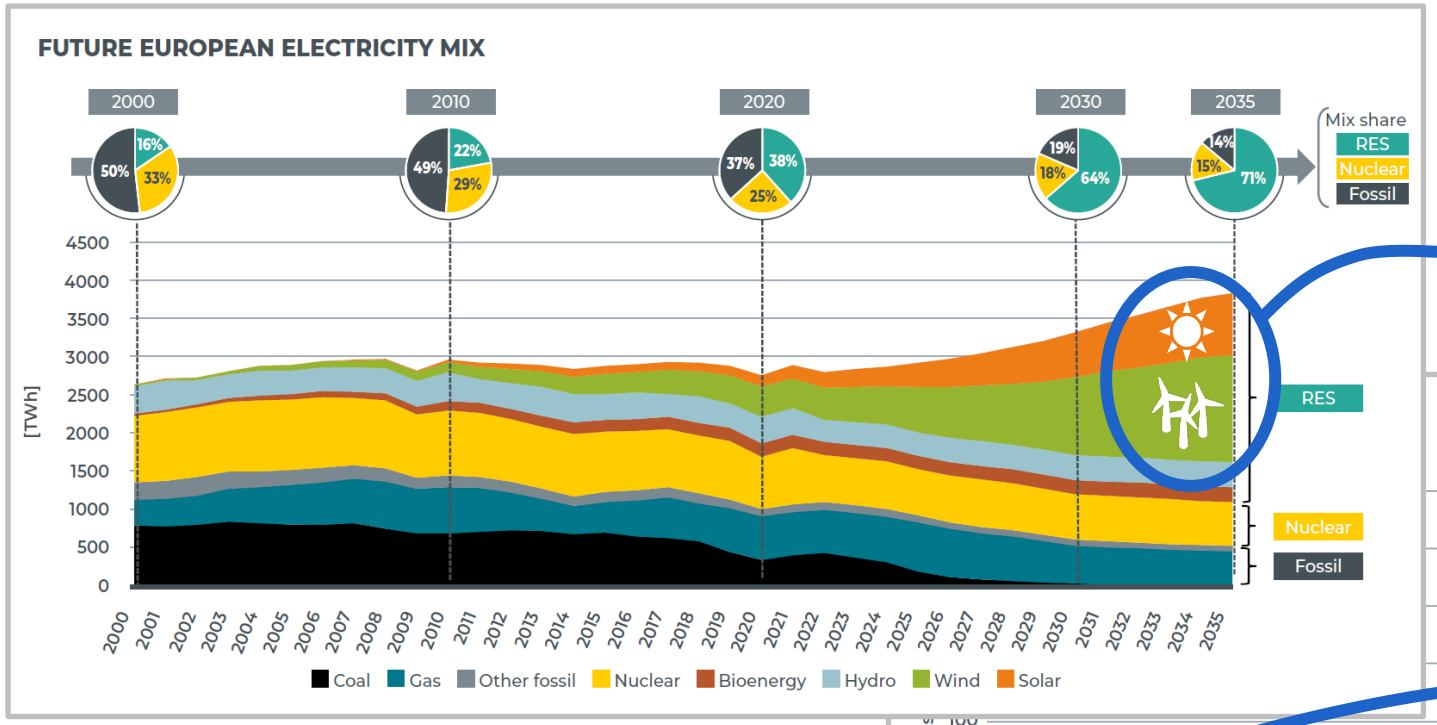
GHENT UNIVERSITY

imec

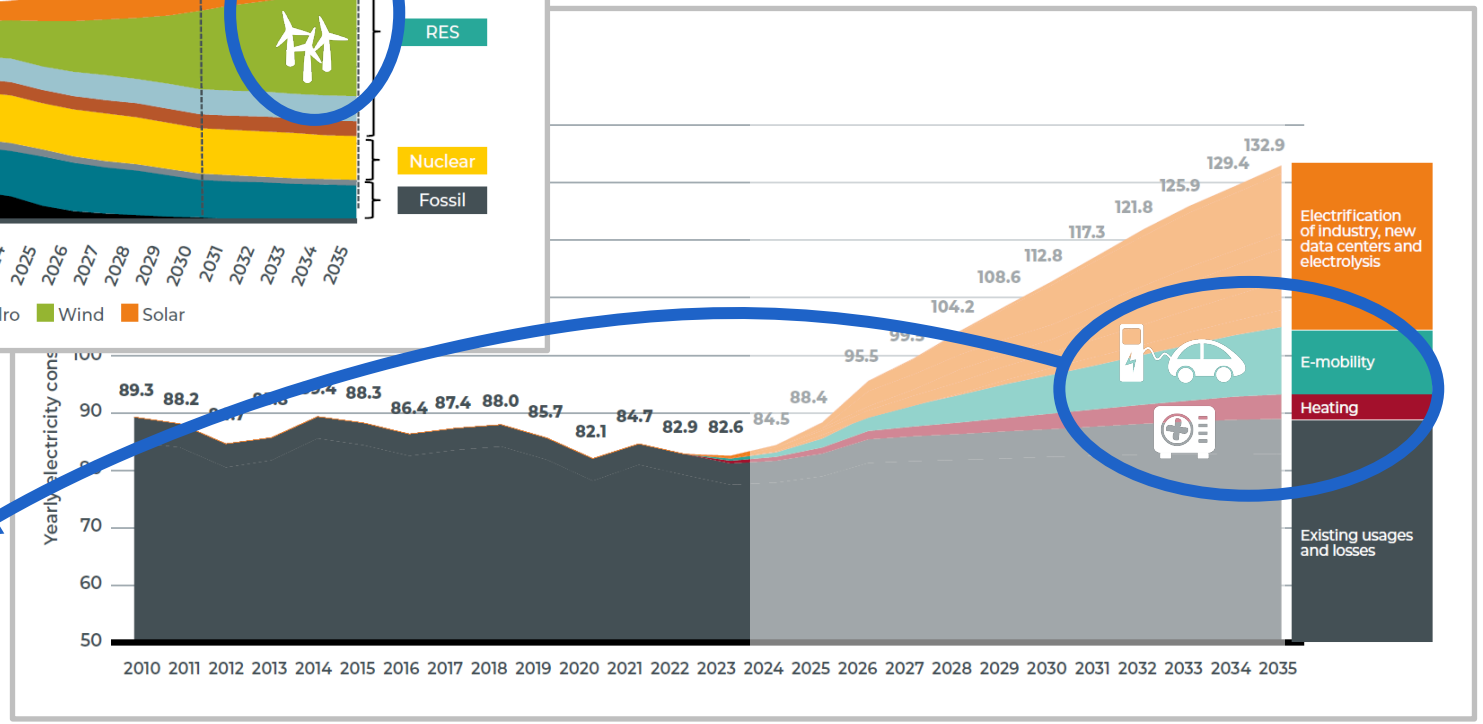
IDLab
INTERNET & DATA LAB

Energy Transition

Source: Adequacy & flexibility study for Belgium (2024-2034), ELIA



RES share in EU electricity supply: ~45% by 2035



e-Mobility + Heat pumps in BE electricity demand: ~25% by 2035

Source: Adequacy & flexibility study for Belgium (2024-2034), ELIA

AI4E: VISION

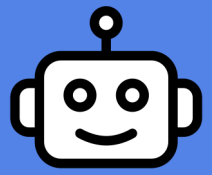


Research, develop and incubate data-driven (decision making) technology supporting **decarbonization of our society** through co-creation with industry & research partners

AI4E: FIT-FOR-PURPOSE & HOLISTIC SOLUTIONS

Neural Networks

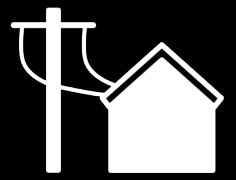
Reinforcement Learning



Graph Neural Networks

. . .

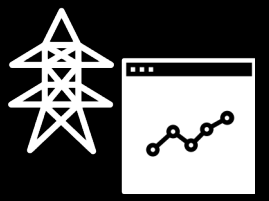
Data-efficient
Control

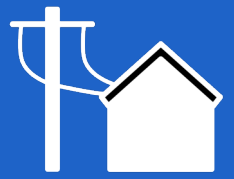


Scalable
Flexibility Exploitation



Predictive
Support Tools





Home Energy Management

Data-efficient learning to control: Physics-informed RL for residential heat pumps

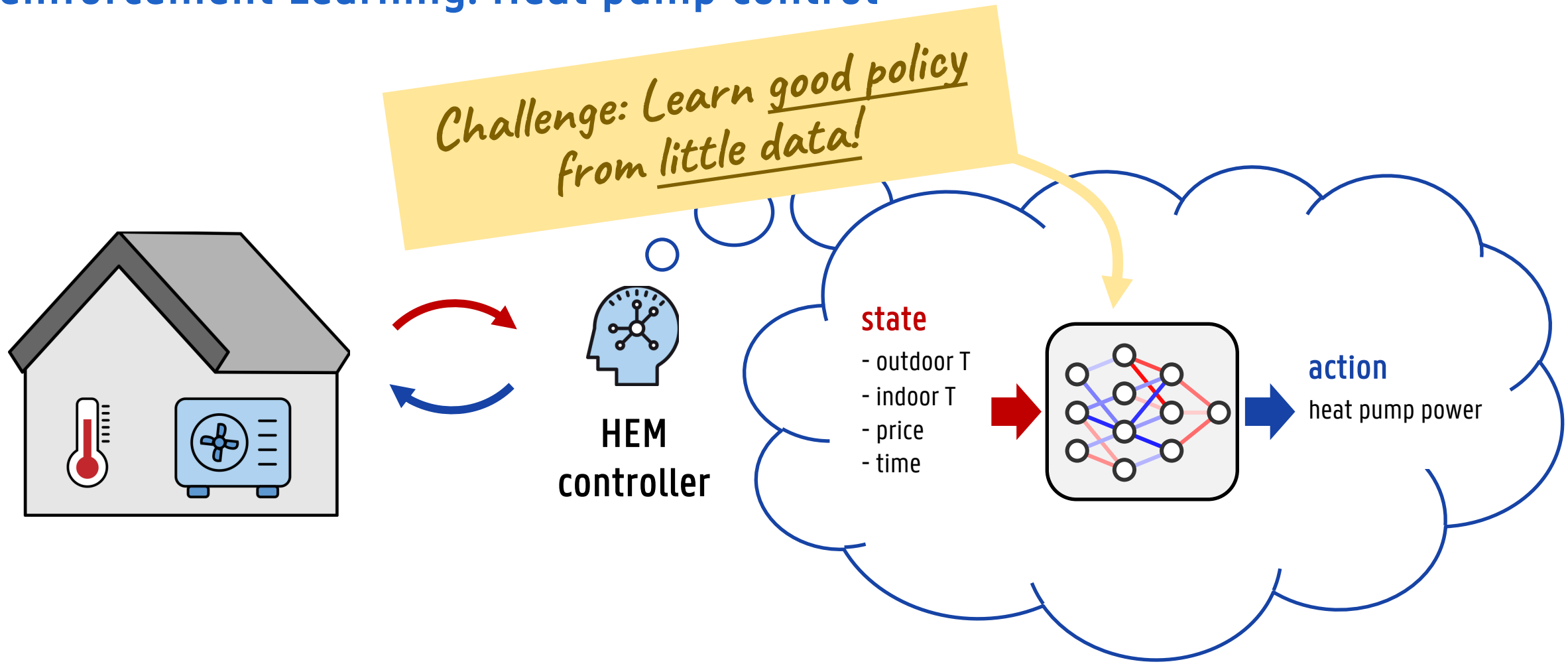
G. Gokhale, B. Claessens, and C. Develder, **"Physics-informed neural networks for control-oriented thermal modeling of buildings"**, Appl. Energy, Vol. 314, 15 May 2022, pp. 1-10. doi:10.1016/j.apenergy.2022.118852

G. Gokhale, B. Claessens, and C. Develder, **"PhysQ: A physics-informed reinforcement learning framework for building controls"**, Arxiv preprint, 21 Nov. 2022, arXiv:2211.11830v1

Thanks Gargya!



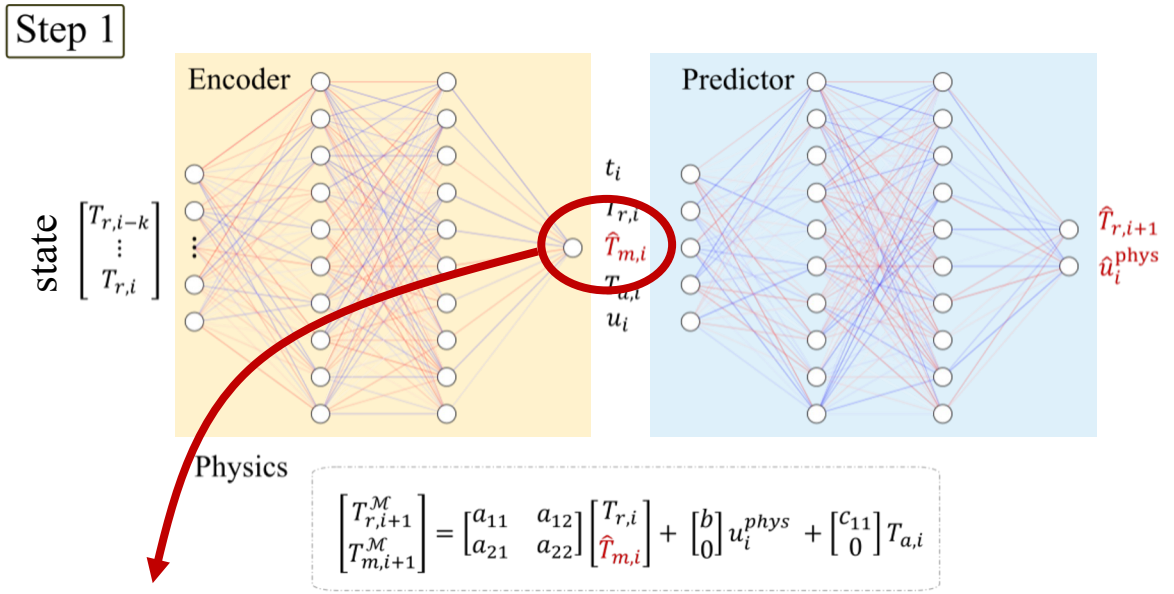
Reinforcement Learning: Heat pump control



PhysQ: A physics-informed reinforcement learning framework

- **PhysQ**:
  - Physics informed RL framework
  - Leverages prior physics to learn physically relevant **latent representations**
  - Learns a control policy using this latent representation

- Two-step Algorithm:
  - **Step 1**: Learn physically relevant **representation** (bulk temperature T_m), from predicting state transitions
  - **Step 2**: Learn a low-dimensional **Q-function** based on this representation, to perform the actual control



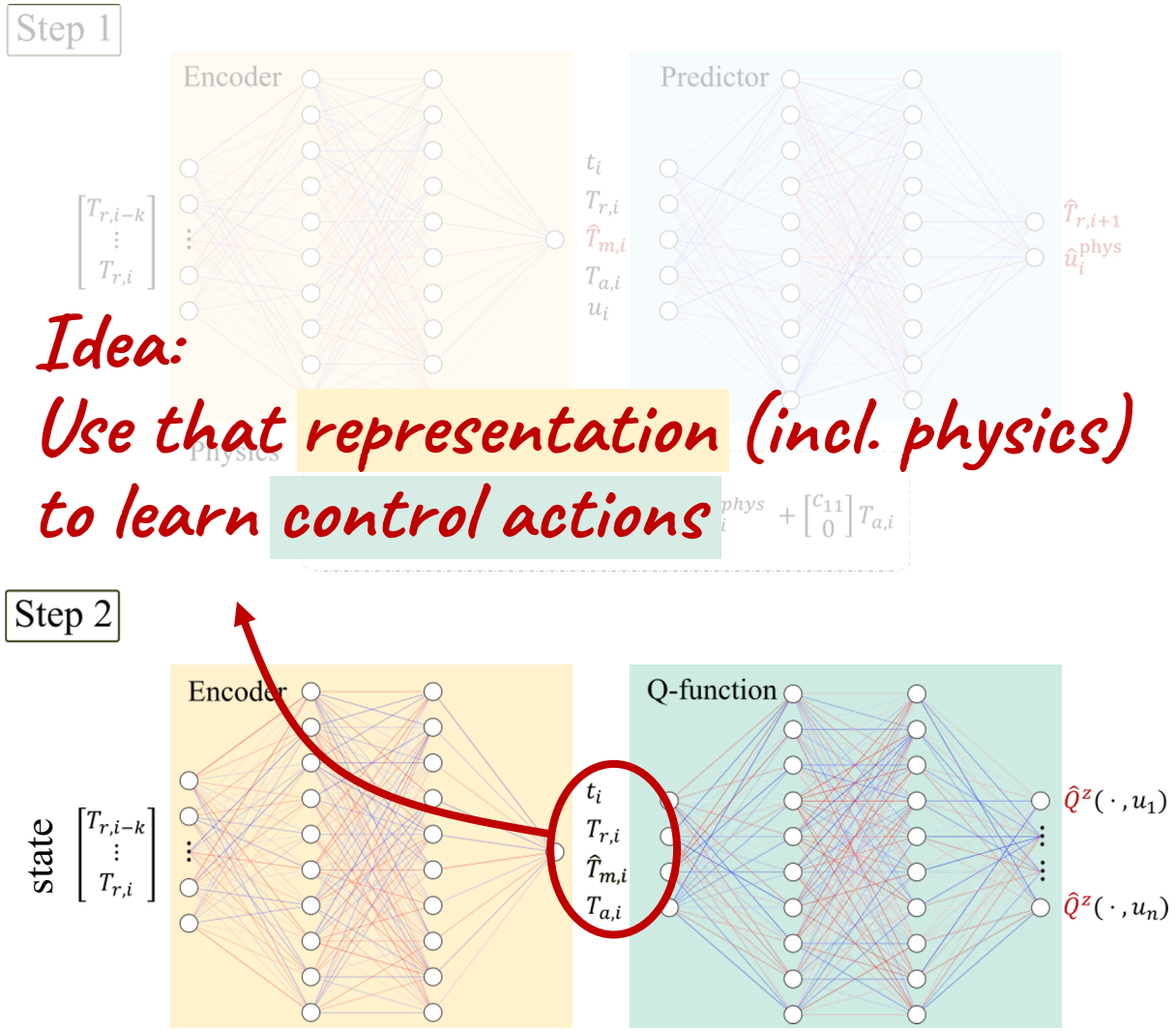
Idea:

- *Add physical quantity (following laws of physics) to representation*
- *Here: building mass temperature; learn it from indoor temp. prediction*

PhysQ: A physics-informed reinforcement learning framework

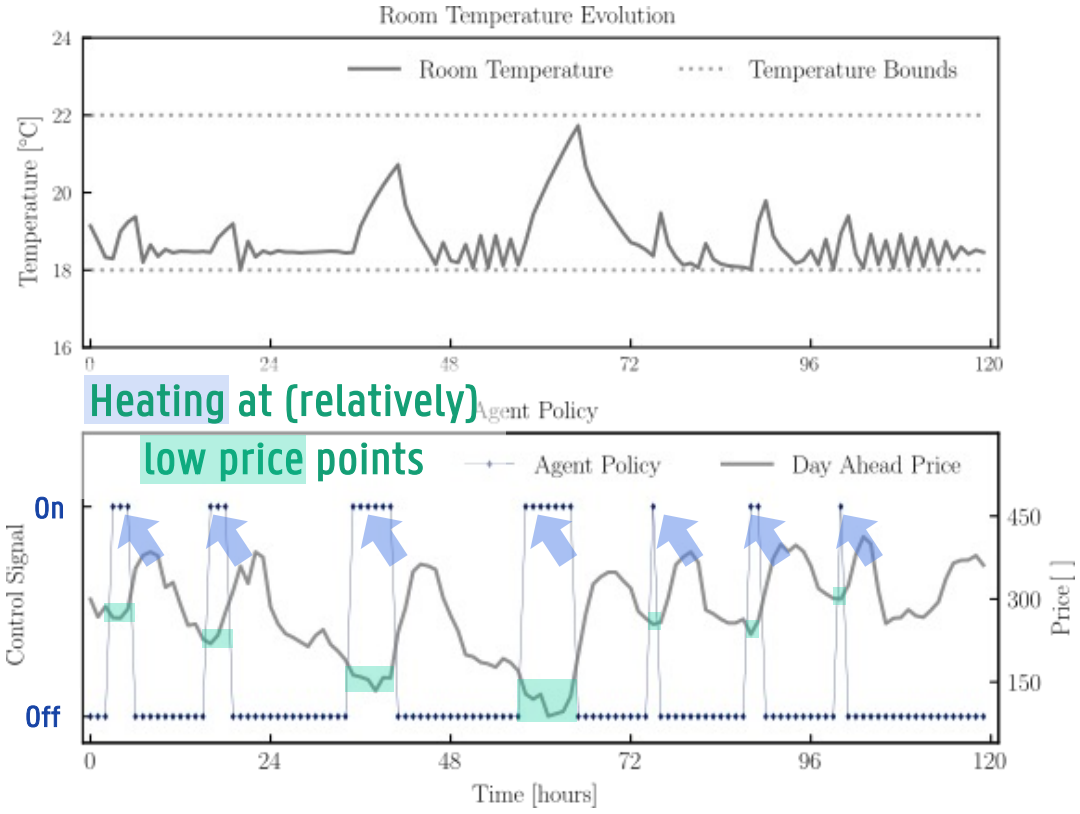
- **PhysQ**:
  - Physics informed RL framework
  - Leverages prior physics to learn physically relevant **latent representations**
  - Learns a control policy using this latent representation

- Two-step Algorithm:
  - **Step 1**: Learn physically relevant **representation** (bulk temperature T_m), from predicting state transitions
  - **Step 2**: Learn a low-dimensional **Q-function** based on this representation, to perform the actual control



Simulation results

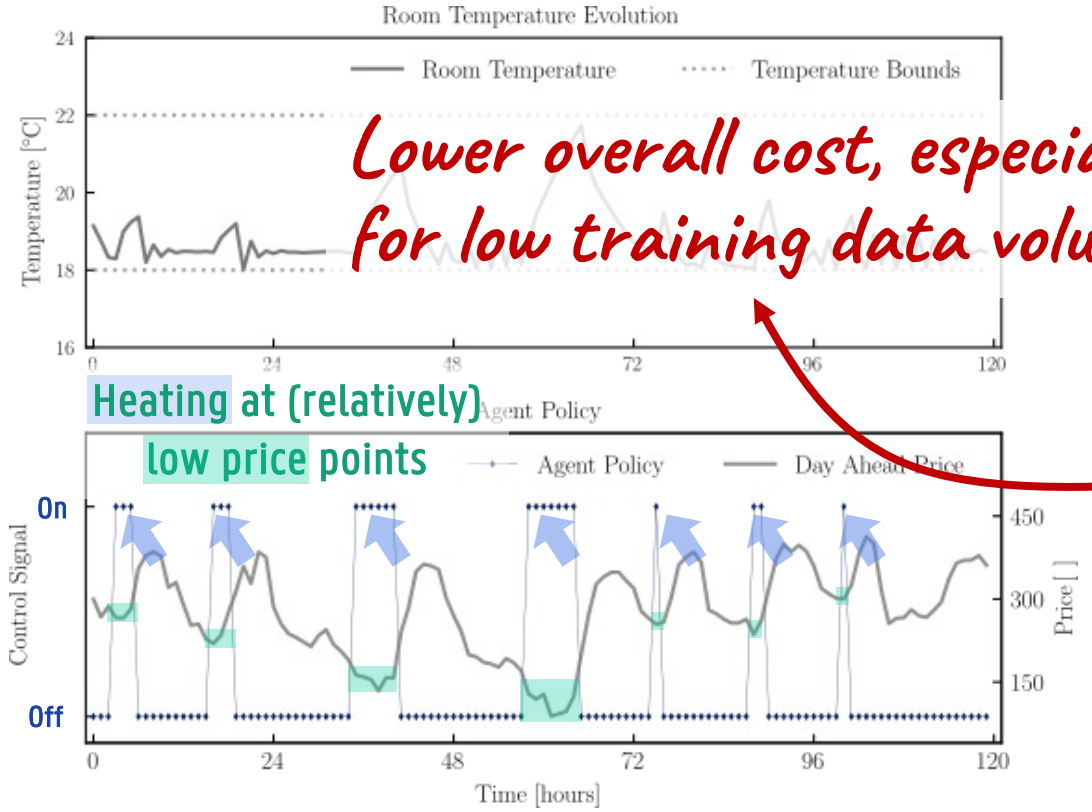
Results for BELPEX prices using a building simulator

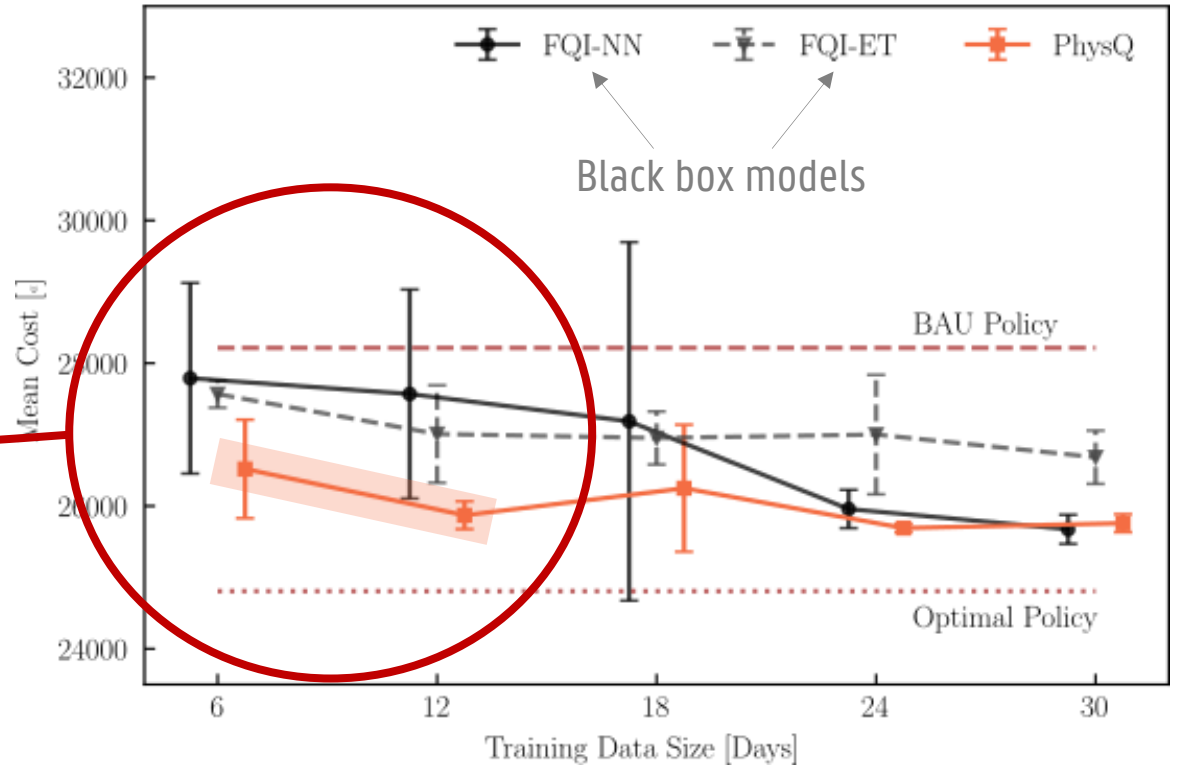


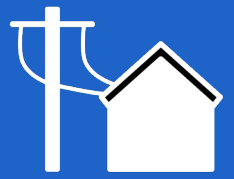
Simulation results

Results for BELPEX prices using a building simulator

Comparison over different training data sizes and agent types







Home Energy Management

Interpretable decision trees distilled from black-box RL policies

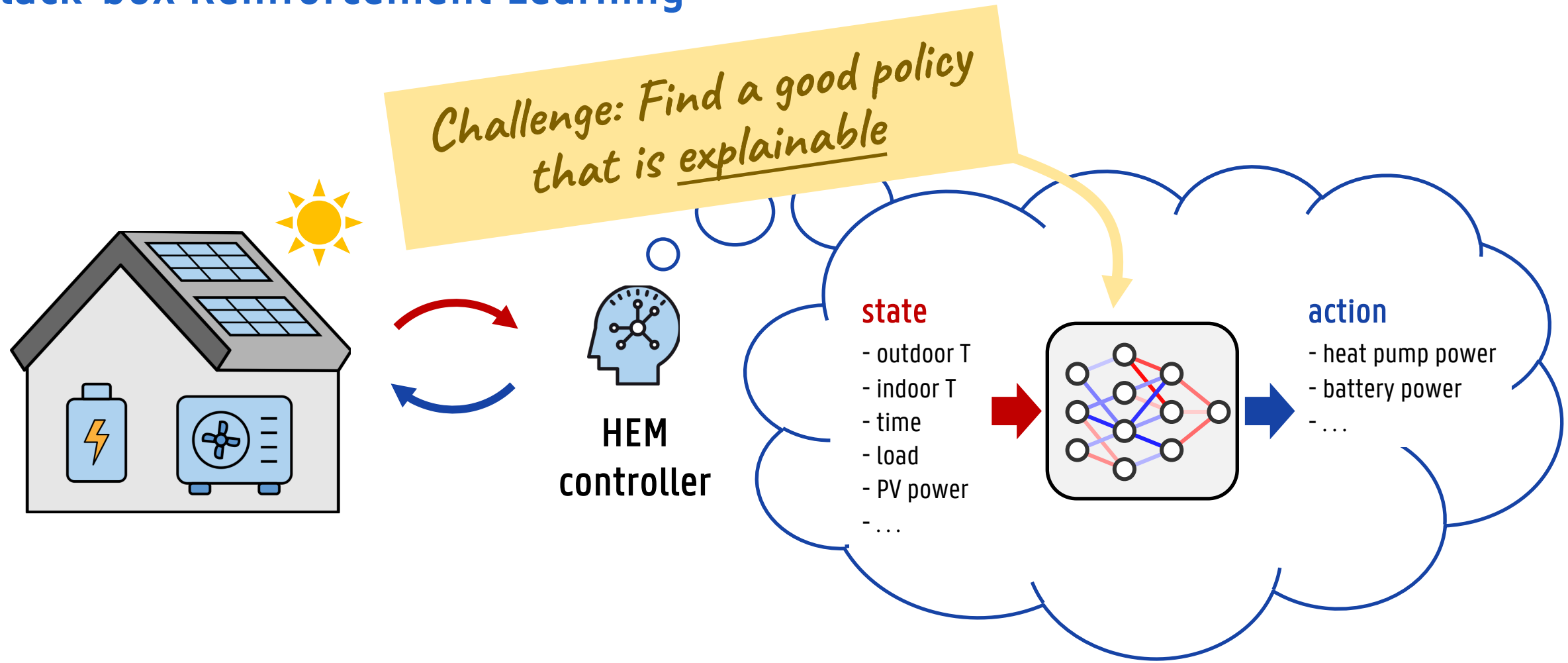
G. Gokhale, S.S. Karimi Madahi, B. Claessens and C. Develder, **"Distill2Explain: Differentiable decision trees for explainable reinforcement learning in energy application controllers"**, in Proc. 15th ACM Int. Conf. Future Energy Sys. (e-Energy 2024), Singapore, 4-7 Jun. 2024, pp. 1-8. doi:10.1145/3632775.3661937

T. Van Puyvelde, M. Zareh and C. Develder, **"Interpretable reinforcement learning for heat pump control through asymmetric differentiable decision trees"**, in Proc. AI-Driven Energy Effic. Dyn. Syst. (AI-DEEDS 2025) at ACM e-Energy 2025, Rotterdam, The Netherlands, 17 Jun. 2025, pp. 968-972. doi:10.1145/3679240.3734671

Thanks Gargya & Toon!

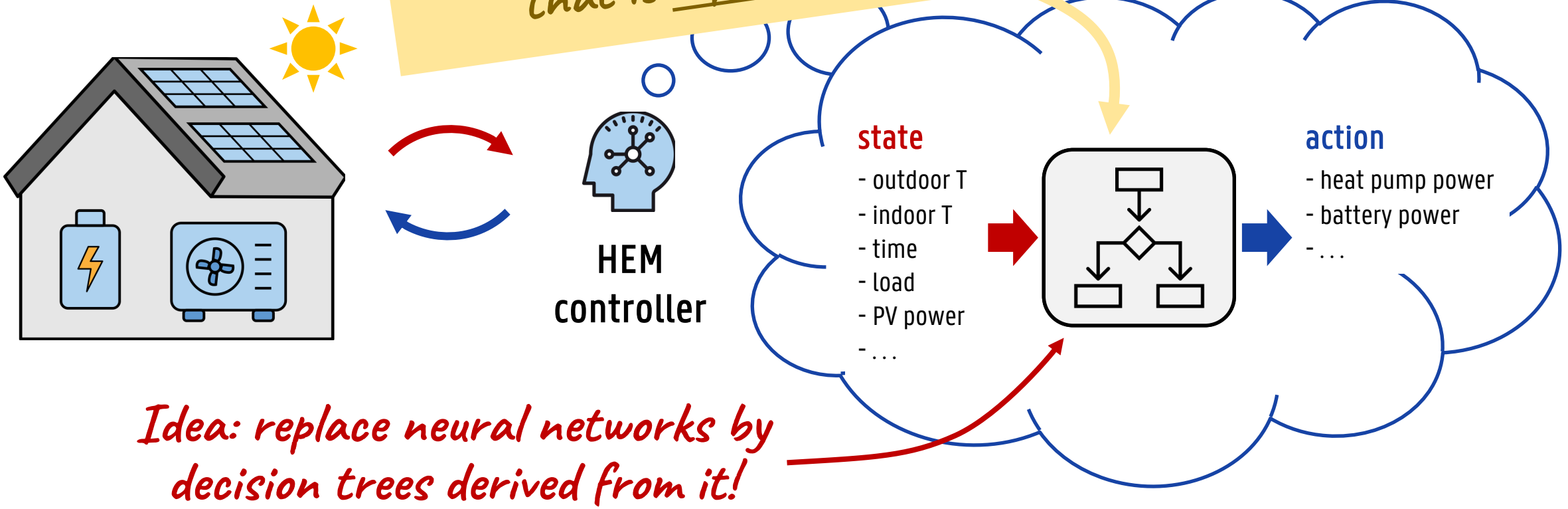


Black-box Reinforcement Learning



Black-box Reinforcement Learning

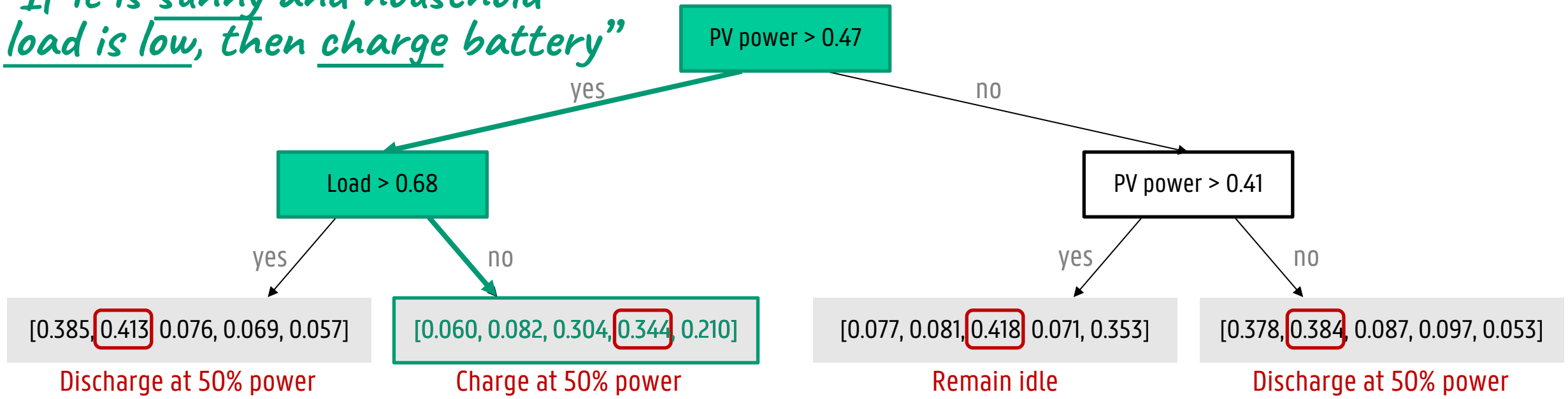
Challenge: Find a good policy that is explainable



Idea: replace neural networks by decision trees derived from it!

Enhancing explainability: Differentiable Decision Trees

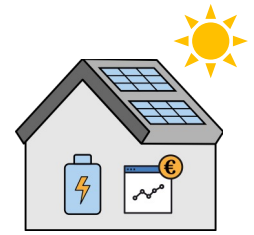
"If it is sunny and household load is low, then charge battery"

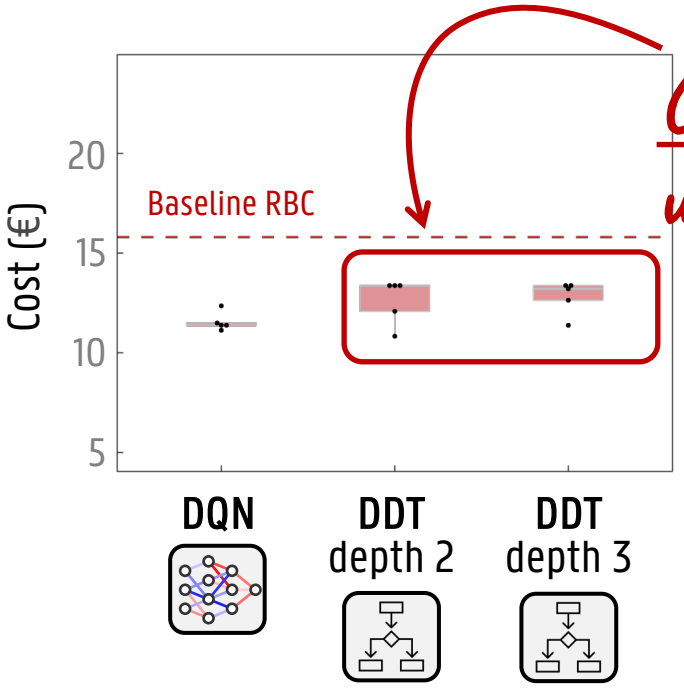


Action probabilities: [discharge at 100% power , discharge at 50% power , idle , charge at 50% power , charge at 100% power]

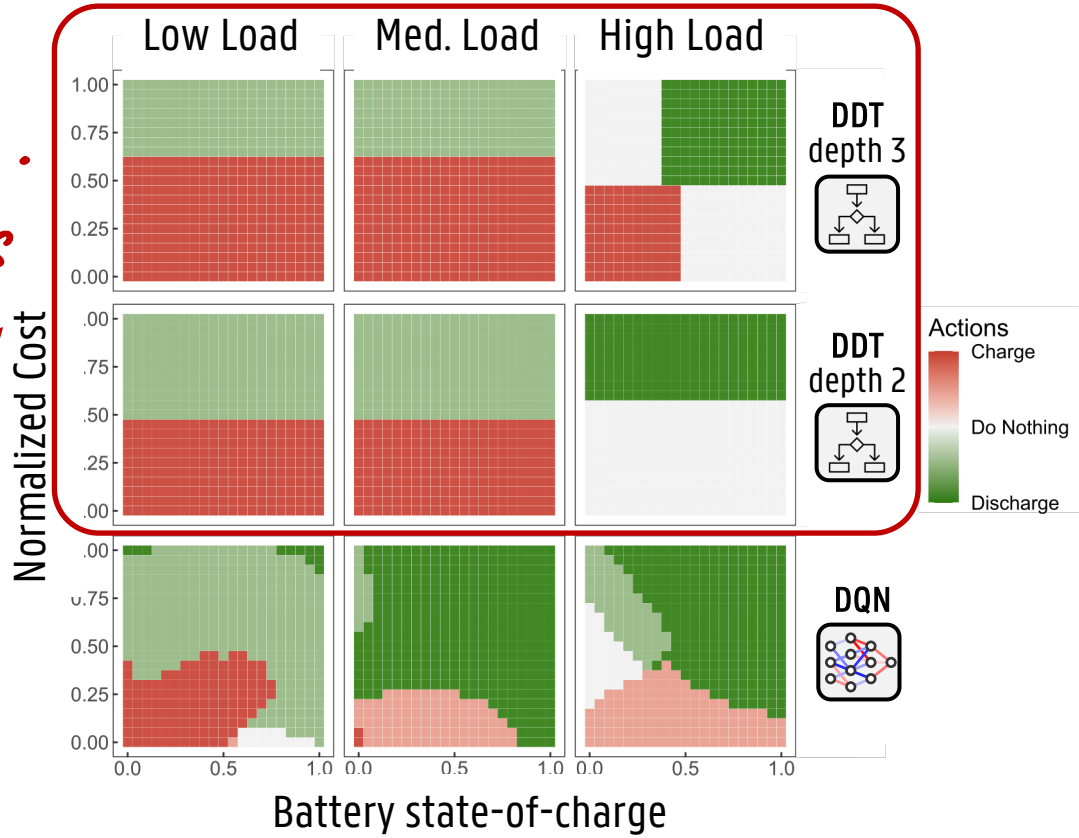
Sample result: Home battery control under time-varying pricing

- Household with rooftop PV and battery
- Day-ahead pricing (qh based) and capacity tariff (= based on max. peak load)



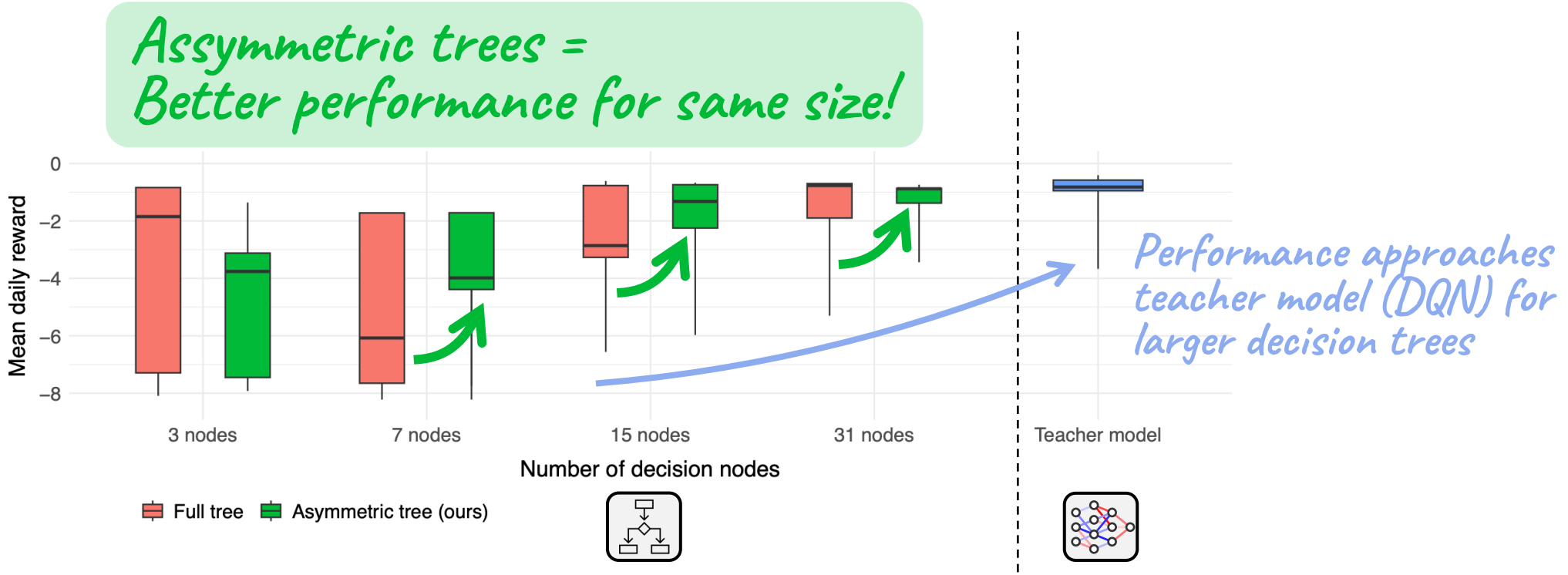


Comparable cost . . .
with simpler policies



Sample result: Heat-pump control

- Same idea applied to space heating with a heat pump (= minimization of cost while maintaining comfort)
- Extended methodology to allow asymmetric trees (= branches can have different depths)



Flexibiity
Optimization

Scalable Attention-based RL for Multi-asset Control

S.S. Karimi Madahi, G. Gabriele, B. Claessens and C. Develder, **"Scalable attention-based reinforcement learning method for multi-asset control"**, in Proc. Workshop Comput. Optim. Buildings (CO-BUILD) at ICML 2025, Vancouver, Canada, 18 Jul. 2025, pp. 1-8.. https://openreview.net/attachment?id=3h0v1Ht73L&name=pdf

G. Gabriele, F. Pavirani, S.S.K. Madahi, and C. Develder, **"Forecasting what matters: Decision-focused RL for controlled EV charging with unknown departure times"**, in Proc. 17th ACM Int. Conf. Future Energy Sys. (e-Energy 2026), Banff, Canada, 23-25 Jun. 2026, pp. 235-239. doi:10.1145/3744255.3811736

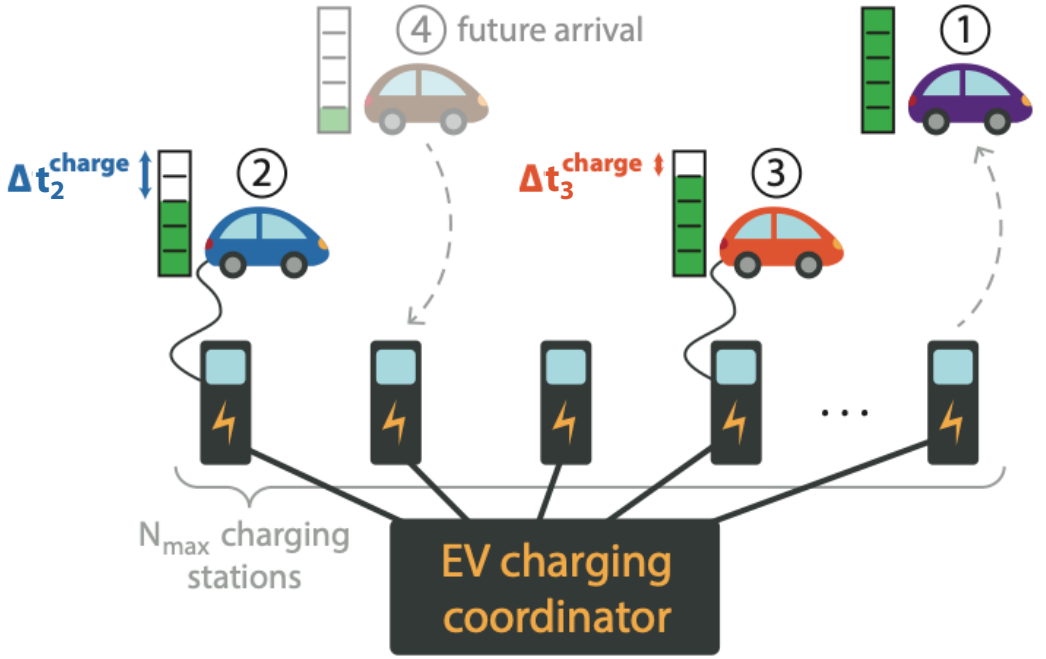
Thanks Soroush & Giuseppe!



Controlling EV charging

- Why?
  - Limit peak load
  - Limit charging costs
  - Balancing
  - …
- How?

AI for EV charging =
(1) KNOWING and
(2) CONTROLLING EV sessions



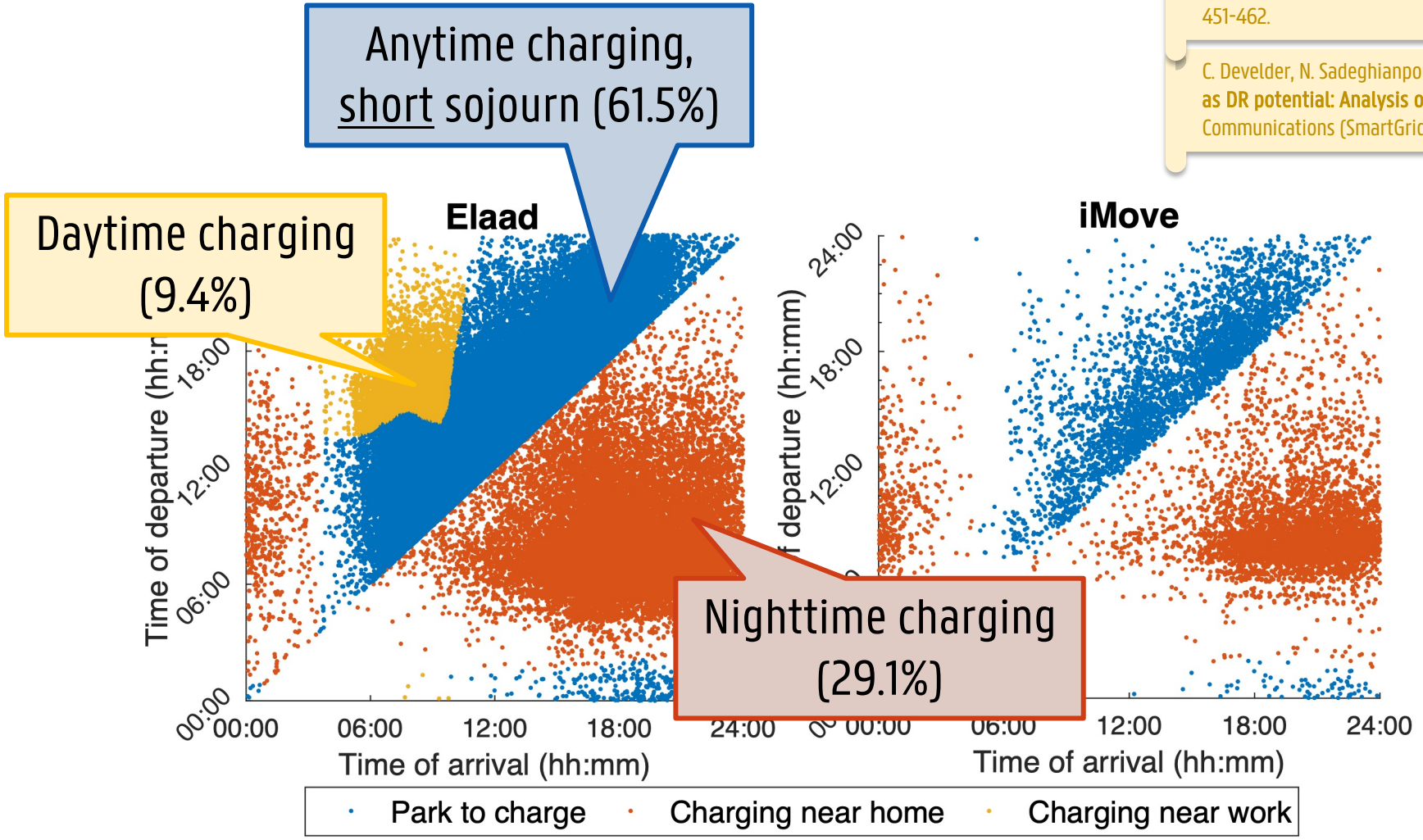
input: arrival & departure times, charging needs
output: which EVs to charge now?

Step 1: Charging session characterization

M. Lahariya, D.F. Benoit and C. Develder, **"Synthetic data generator for electric vehicle charging sessions: Modeling and evaluation using real-world data"**, Energies, Vol. 13, Issue 16, Paper no. 4211, 2020.

N. Sadeghianpourhamami, N. Refa, M. Strobbe and C. Develder, **"Quantitative analysis of electric vehicle flexibility: A data-driven approach"**, Int. J. Electr. Power Energy Syst., Vol. 95, Feb. 2018, pp. 451-462.

C. Develder, N. Sadeghianpourhamami, M. Strobbe and N. Refa, **"Quantifying flexibility in EV charging as DR potential: Analysis of two real-world data sets"**, in Proc. 7th IEEE Int. Conf. Smart Grid Communications (SmartGridComm 2016), Sydney, Australia, 6-9 Nov. 2016, pp. 600-605.

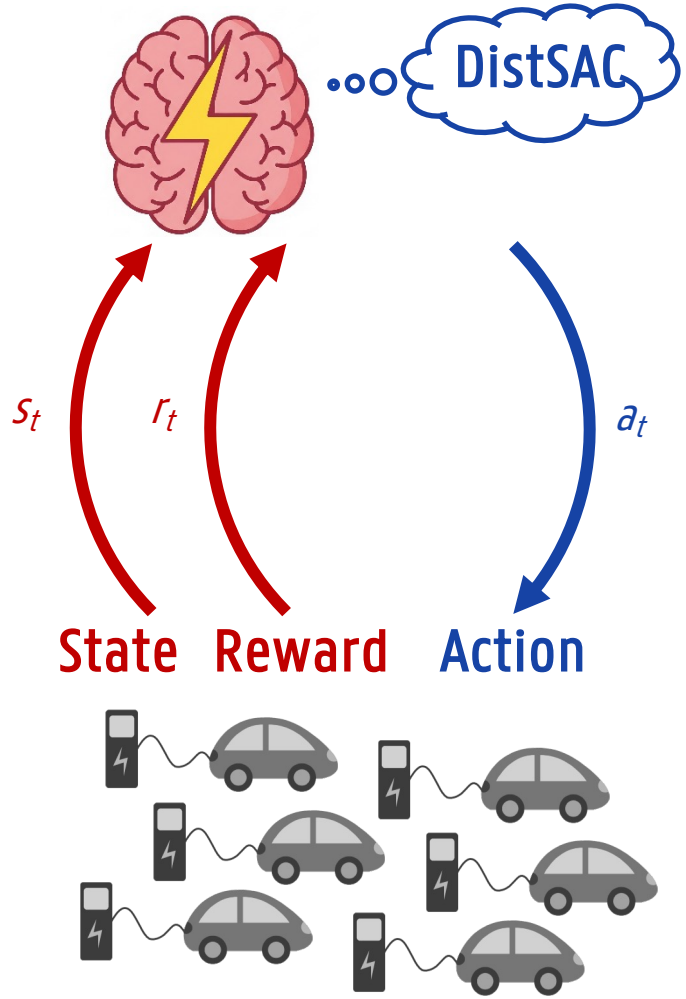




Step 2: Scalable RL controller

- At each timestep t the agent
  - Gets **state** observation $s_t = \bigoplus_{i=1}^{N_t} (\underbrace{t^{\text{arr}}_{i,t}, t^{\text{dep}}_{i,t}, E^{\text{req}}_{i,t}}_{\text{local info}}, \underbrace{t, N_t}_{\text{global info}})$
  - Decides to take **action** $a_t = \bigoplus_{i=1}^{N_t} a_{i,t}, \quad a_{i,t} \in \{0, P^{\max}_i\}$
- The environment
  - Receives action a_t
  - Transitions to **state** s_{t+1}
  - Returns scalar **reward** $r_t = \underbrace{-\left(\sum_{i=1}^{N_t} a_{i,t}\right)^2}_{\text{load flattening}} \underbrace{- \alpha \sum_{i=1}^{N_t} \mathbb{1}\left\{E^{\text{req}}_{i,t} > t^{\text{dep}}_{i,t} \cdot P^{\max}_i\right\}}_{\text{penalty for unfinished charging}}$

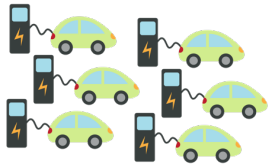
Goal: Minimize peak load + Charge EVs on time



Step 2: Attention-based state representations for scalable RL

Idea:

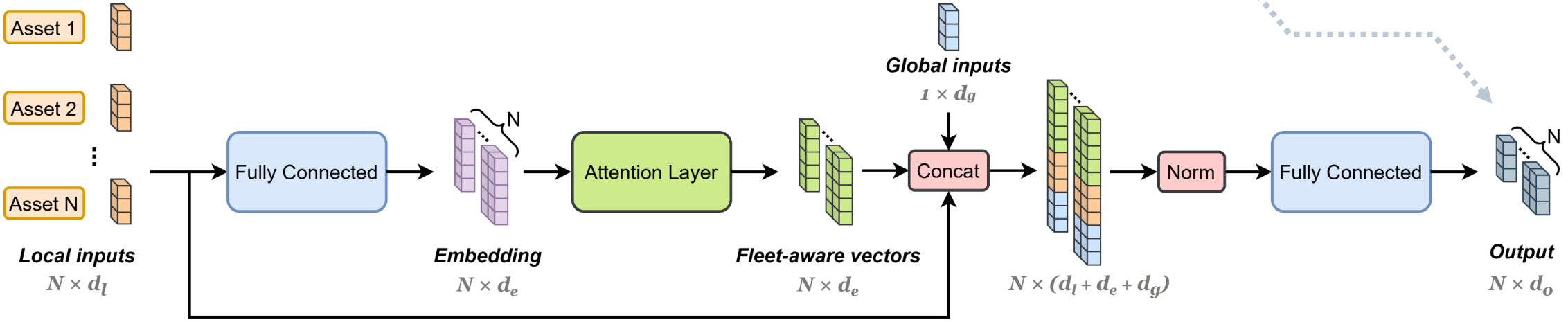






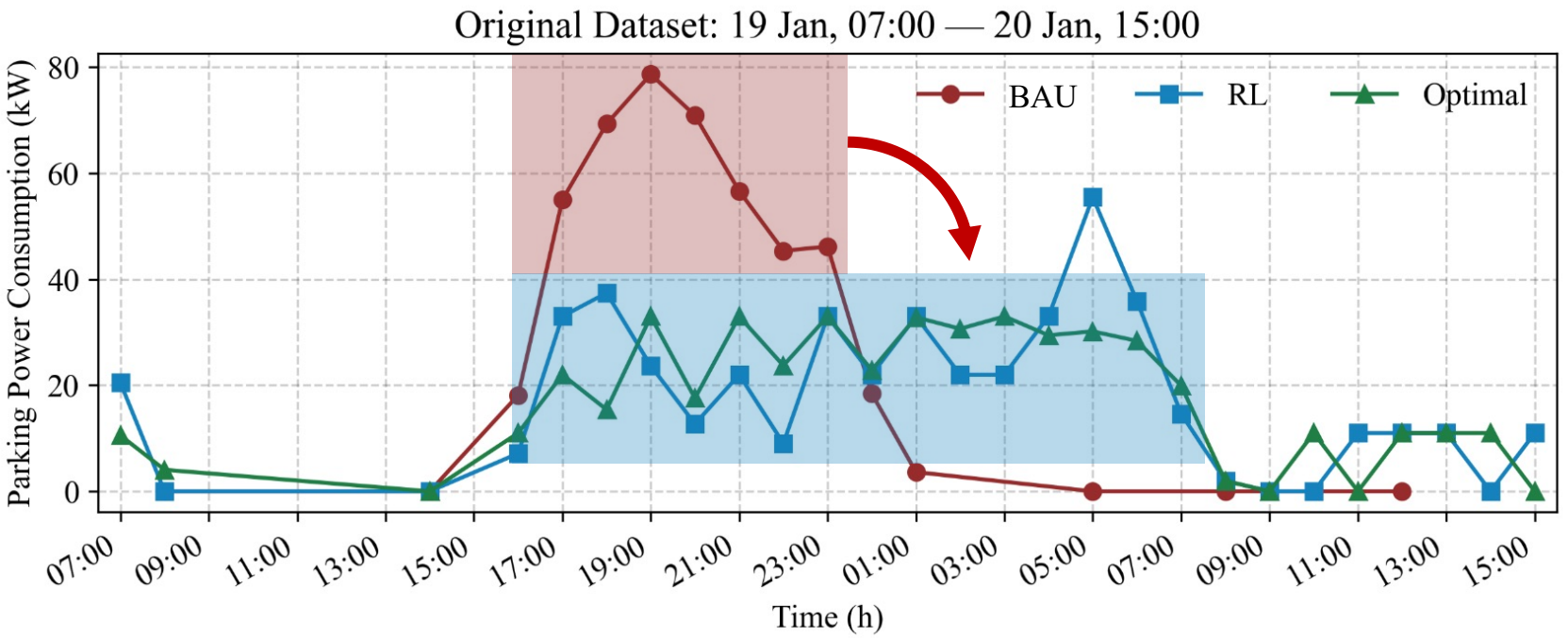
N

- Construct state representation combining own asset + relative to other assets + global state
- Use state representation as input representation in conventional RL (e.g., to map to Q-value, action, …)



Simulation Results

Significant peak reduction compared to uncontrolled charging



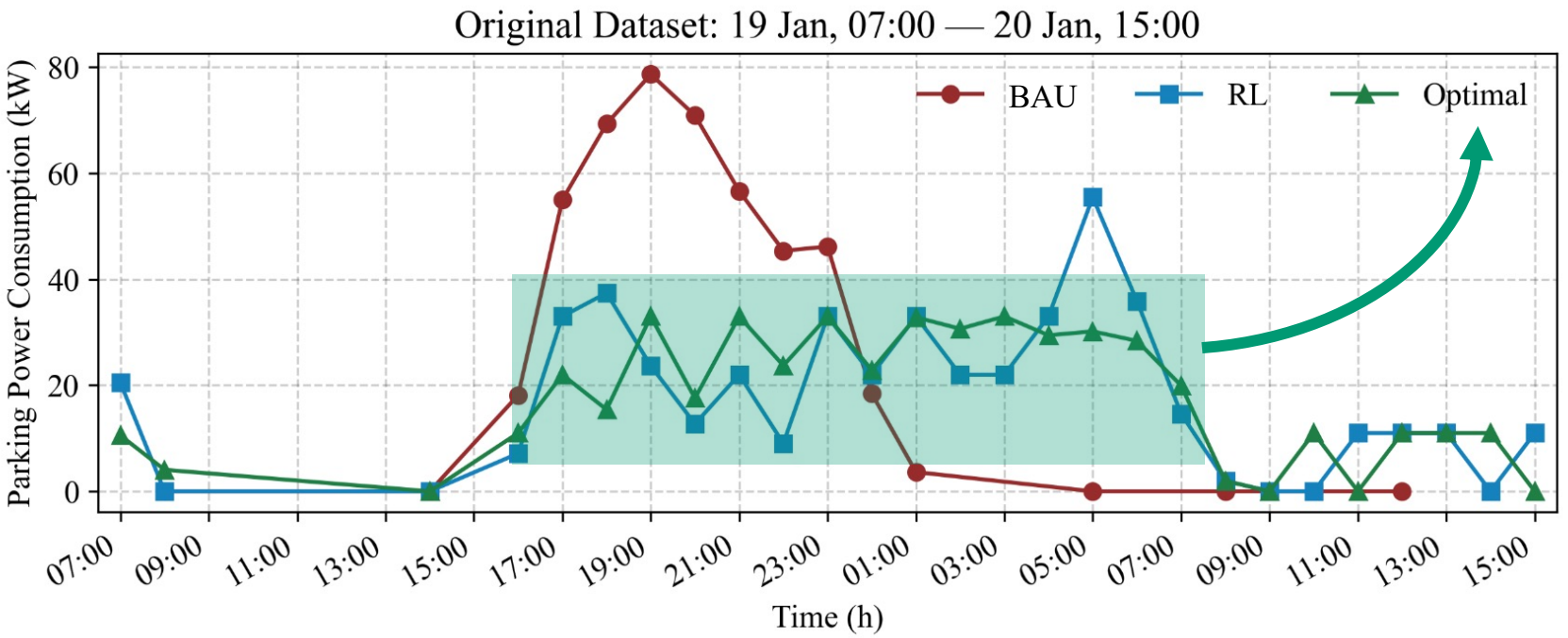
Attention-based RL

- Flattens load (peak reduced by 27%)*
- Performs close to optimization based on perfect foresight

*: 27% is the average over full test set

Simulation Results

Peak reduction of our RL solution is very close to optimal control (w/ perfect foresight)



Attention-based RL

- Flattens load (peak reduced by 27%)*
- Performs close to optimization based on perfect foresight

*: 27% is the average over full test set

Support Tools
for DSO & TSO

Imbalance price prediction with Monte Carlo Tree Search (MCTS)

F. Pavirani, J. Van Gompel, S.S. Karimi Madahi, B. Claessens and C. Develder, **"Predicting and publishing accurate imbalance prices using Monte Carlo Tree Search"**, Appl. Energy, Vol. 392, No. 125944, 15 Aug. 2025. doi:10.1016/j.apenergy.2025.125944

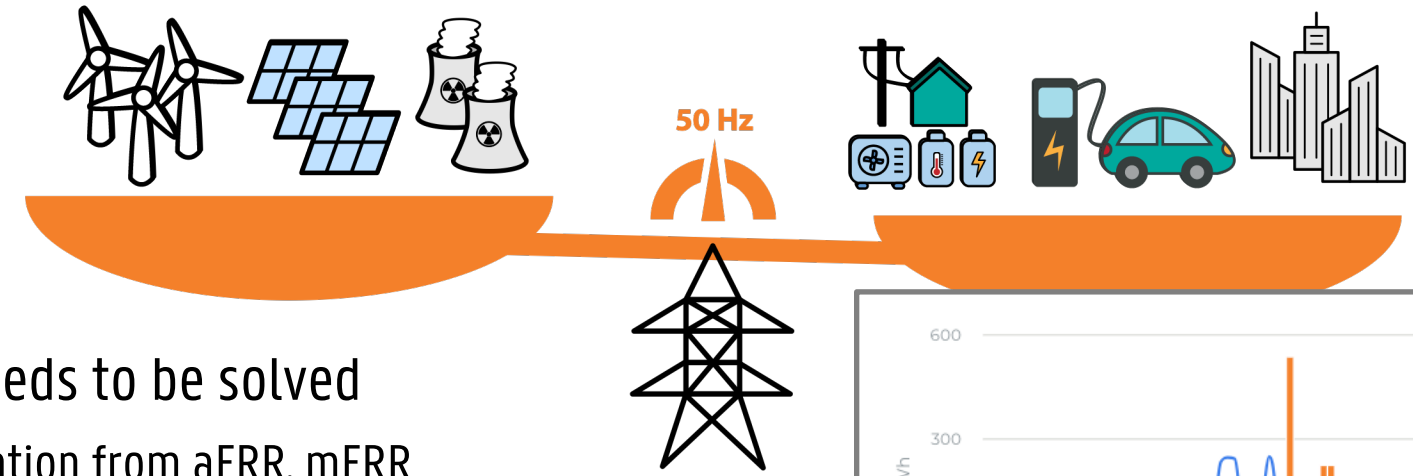
J. Van Gompel, B. Claessens and C. Develder, **"Probabilistic forecasting of power system imbalance using neural network-based ensembles"**, Appl. Energy, Vol. 401, Part B, No. 126714, Dec. 2025. doi:10.1016/j.apenergy.22025.126714

Thanks Fabio & Jonas!



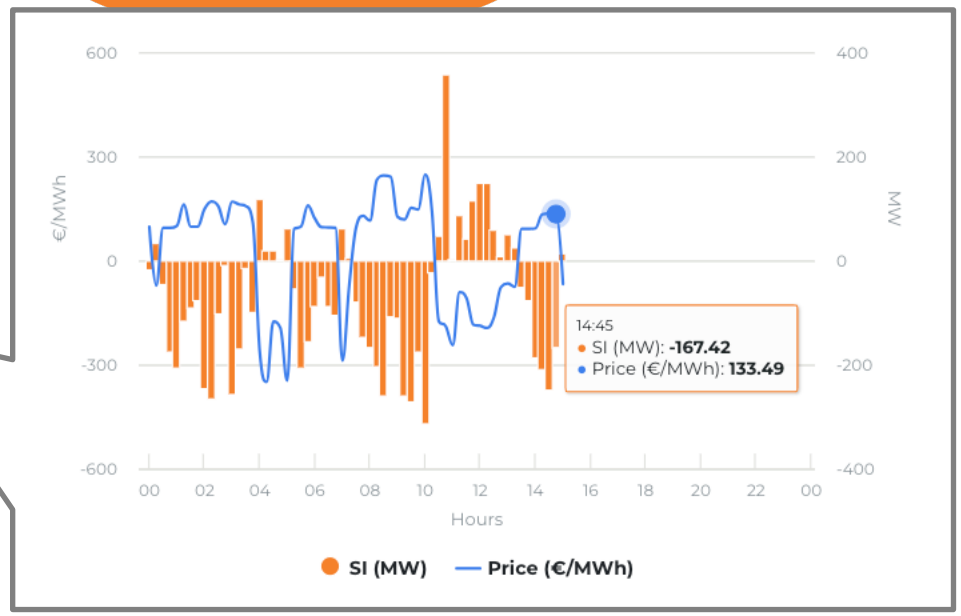
Maintaining System Balance

- **Generation**: Renewable sources ⟷ **Load**: Increased electrification



- Mismatch needs to be solved
  - Compensation from aFRR, mFRR
  - Reaction of BRPs themselves

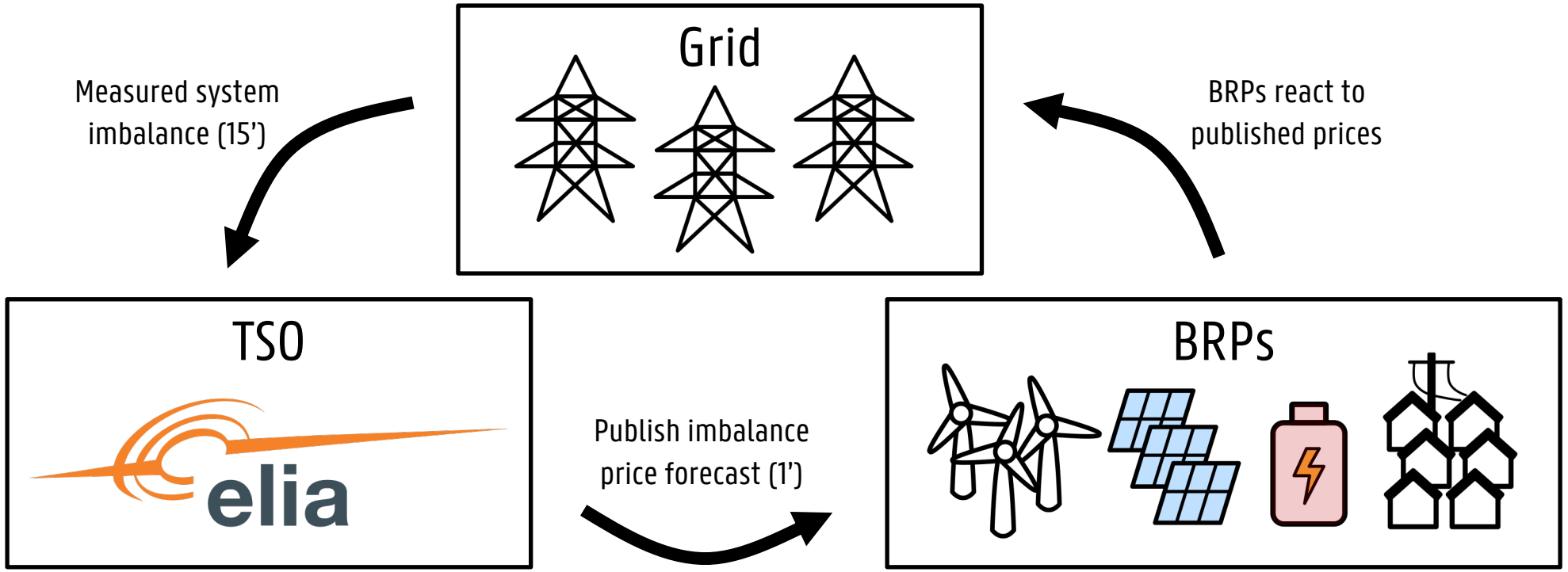
    . . . where imbalance prices for BRP depend on SI



Source: https://www.elia.be/en/grid-data/balancing/imbalance-prices-15-min (Fri, April 2024, 15:00)

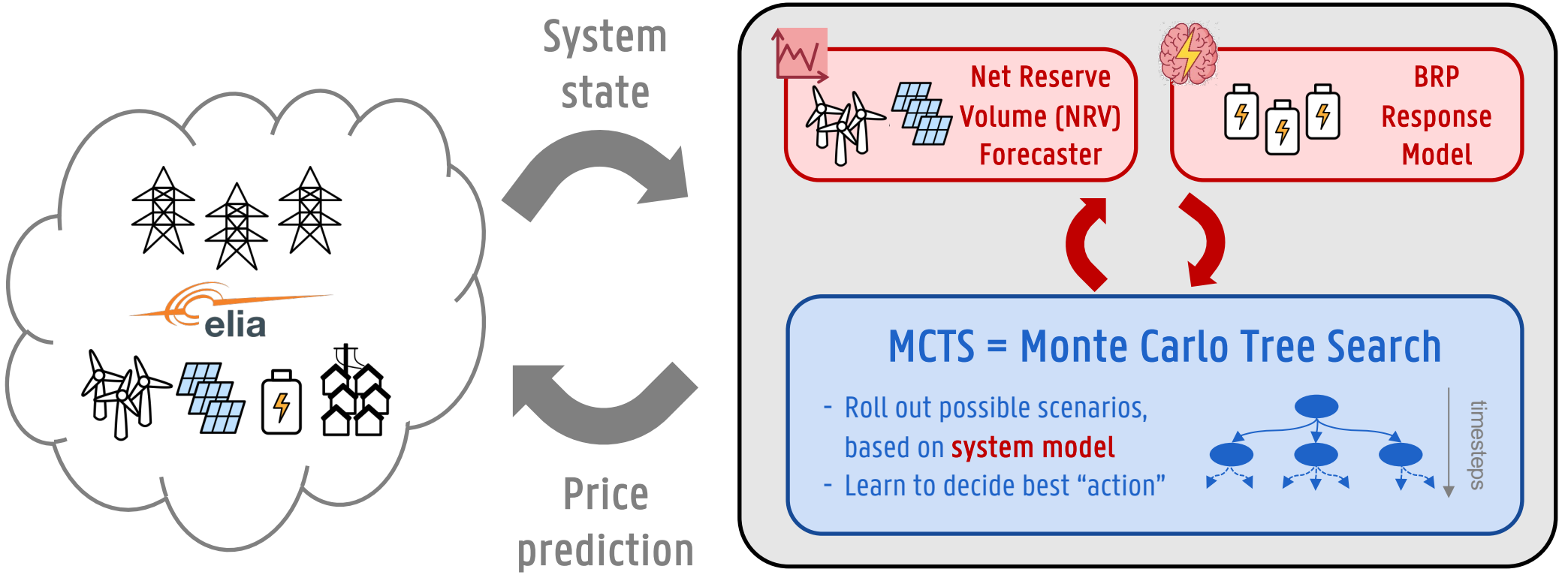
But ... price predictions are published within the quarter! -> influences behavior of BRPs!

Imbalance Price Publication



But ... price predictions are published within the quarter! -> influences behavior of BRPs!

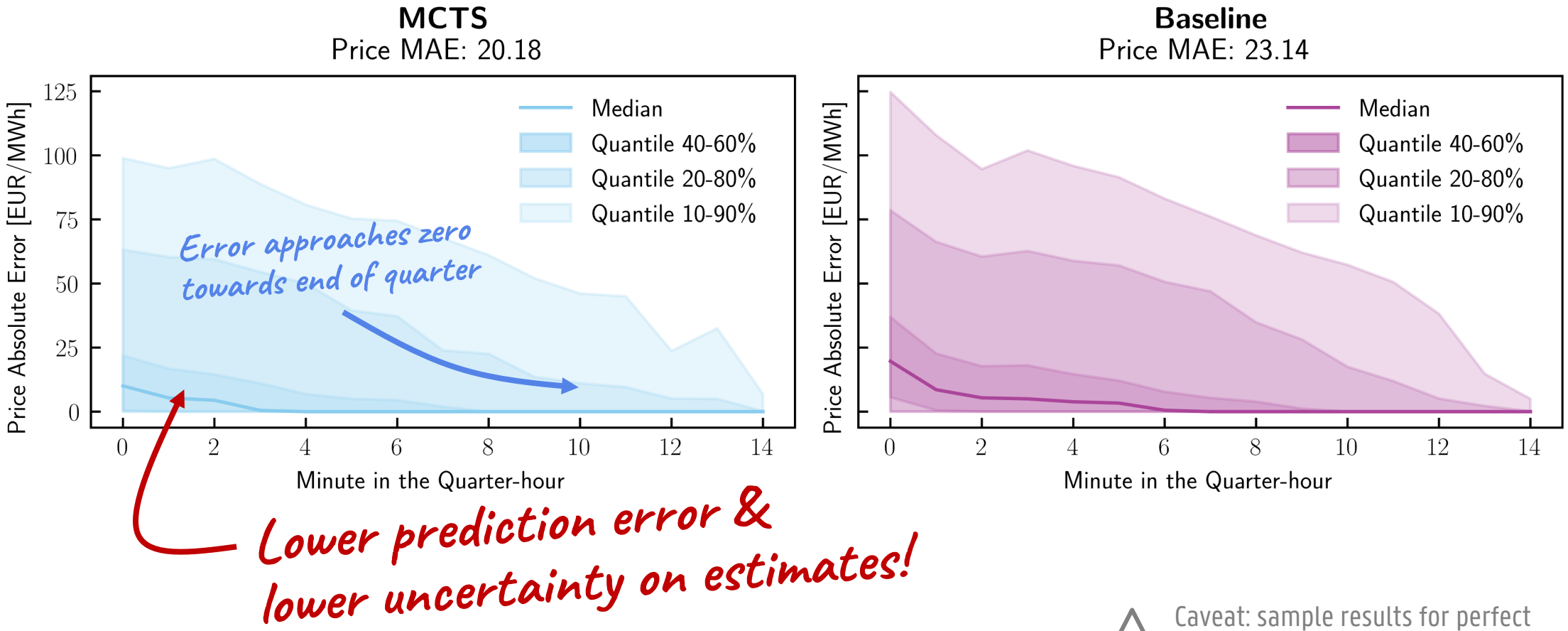
MCTS Based Imbalance Price Forecasting



J. Van Gompel, B. Claessens and C. Develder, **"Probabilistic forecasting of power system imbalance using neural network-based ensembles"**, Appl. Energy, Vol. 401, Part B, No. 126714, Dec. 2025. doi:10.1016/j.apenergy.2025.126714

S.S. Karimi Madahi, B. Claessens and C. Develder, **"Distributional reinforcement learning-based energy arbitrage strategies in imbalance settlement mechanism"**, J. Energy Storage, Vol. 104, Part A, Dec. 2024, pp. 114377. doi:10.1016/j.est.2024.114377

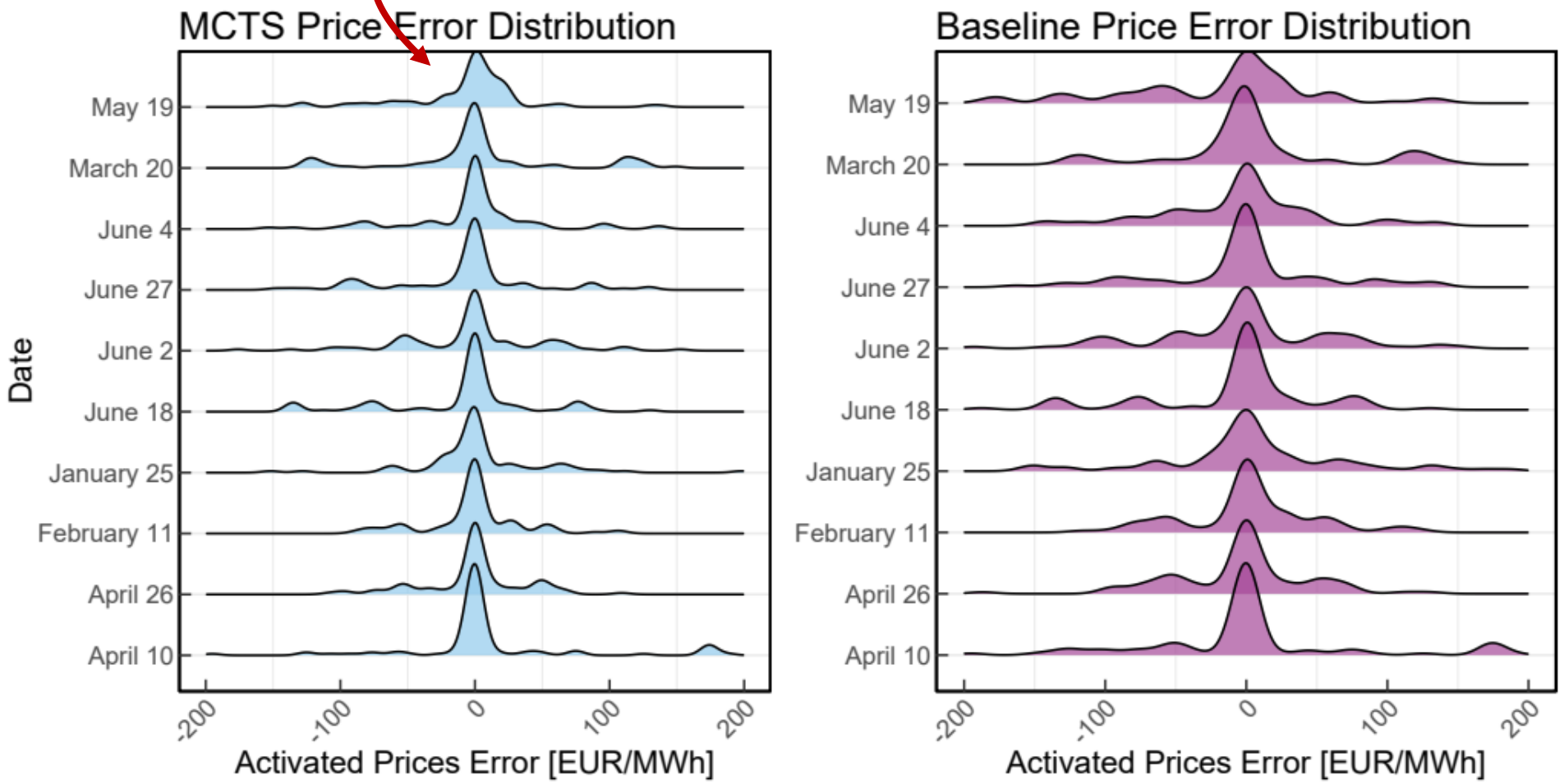
Results: Price prediction within the quarter

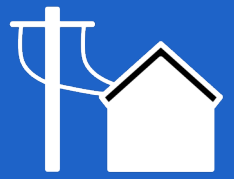


⚠ Caveat: sample results for perfect NRV and BRP response models

Results: Overall error distributions

Distributions of errors are more narrowly centered around zero!





Home Energy Management

A real-world testbed environment: HomeLabGym

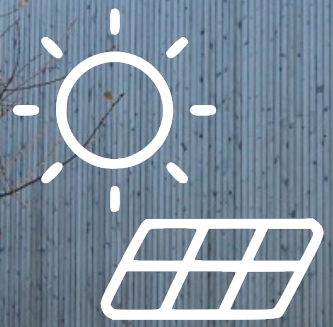
T. Van Puyvelde, M.-S. Verwee, G. Gokhale, M. Zareh Eshdoust and C. Develder, **"HomeLabGym: A real-world testbed for home energy management systems"**, in Proc. 15th ACM Int. Conf. Future Energy Sys. (e-Energy 2024), Singapore, 4-7 Jun. 2024, pp. 1-2. doi:10.1145/3632775.3661974

Thanks Toon!



HomeLab

imec IDLab INTERNET & DATA LAB GHENT UNIVERSITY







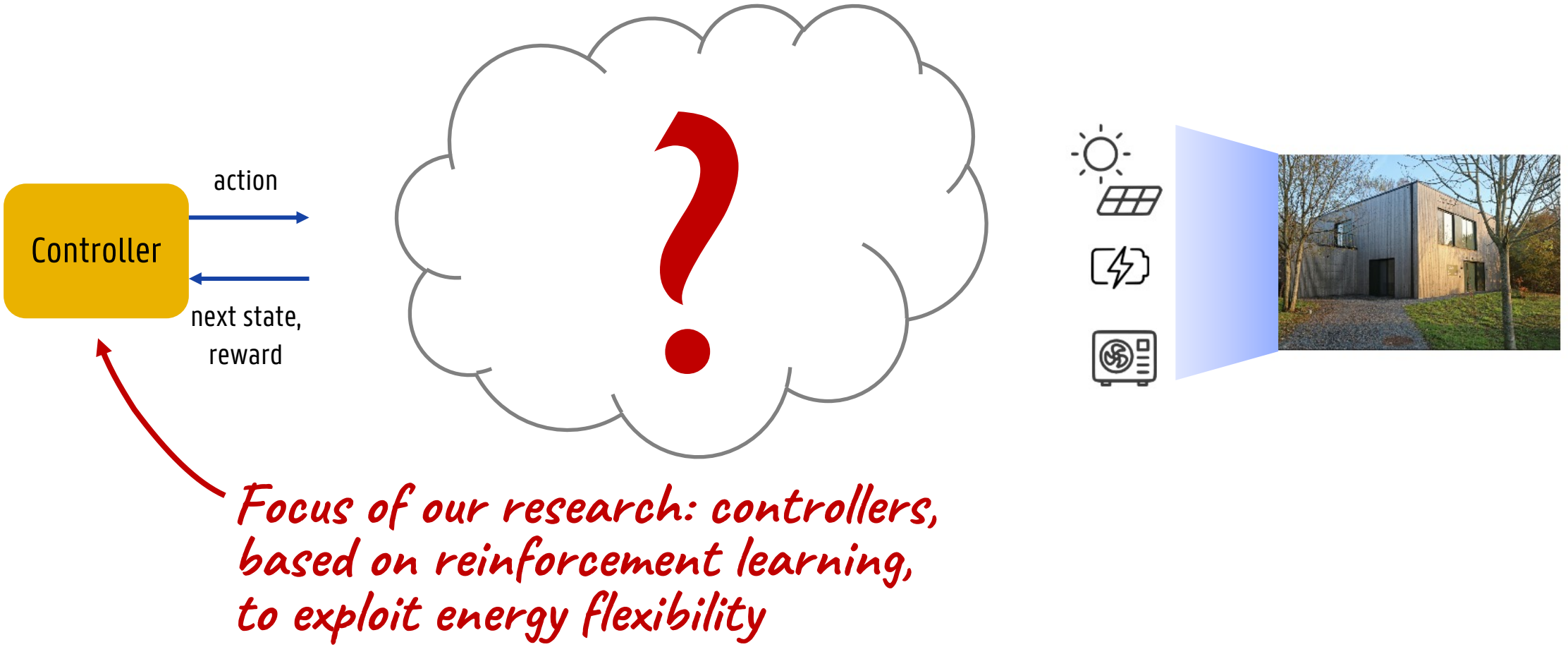
Research infrastructure to validate and optimize IoT & AI innovations in real environment

- >300 sensors (temperature, energy, light, occupancy, …)
- HVAC, ESS, PV
- Controllable blinds and windows
- …



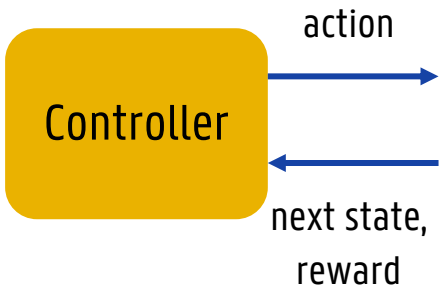
Energy infrastructure

HomeLabGym



Focus of our research: controllers, based on reinforcement learning, to exploit energy flexibility

HomeLabGym

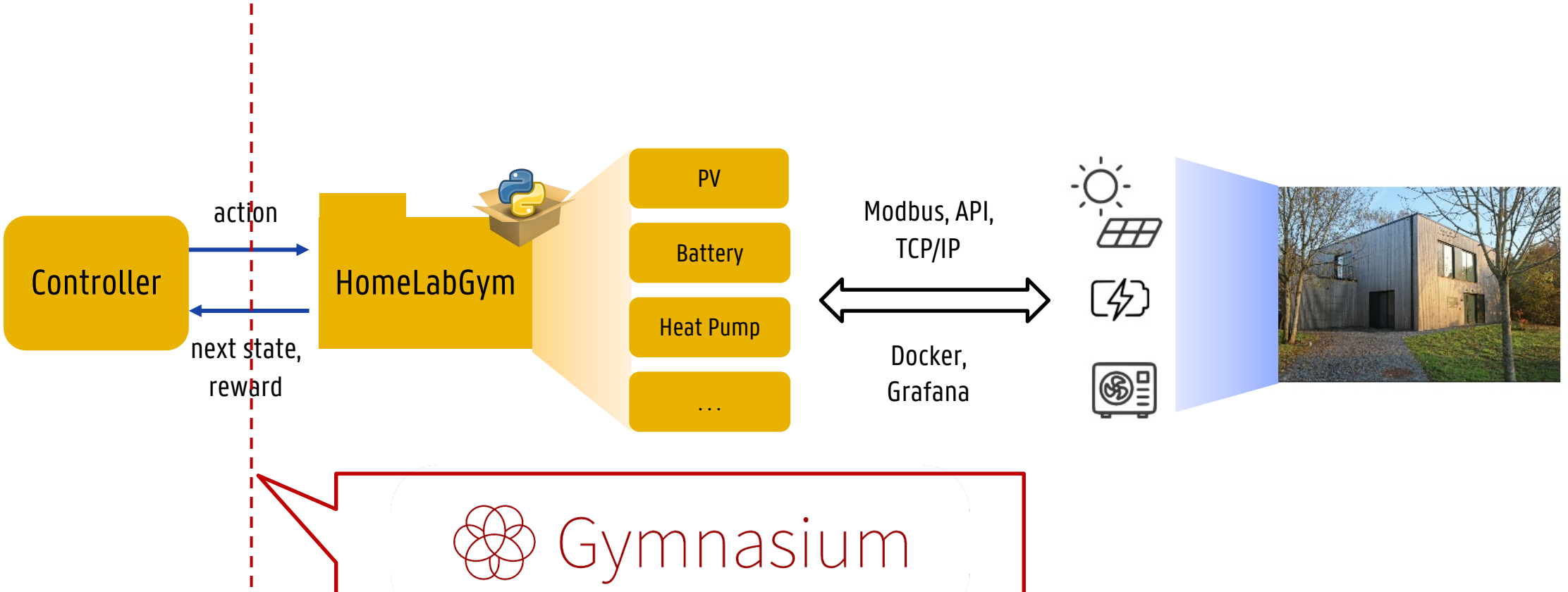


Wishlist:

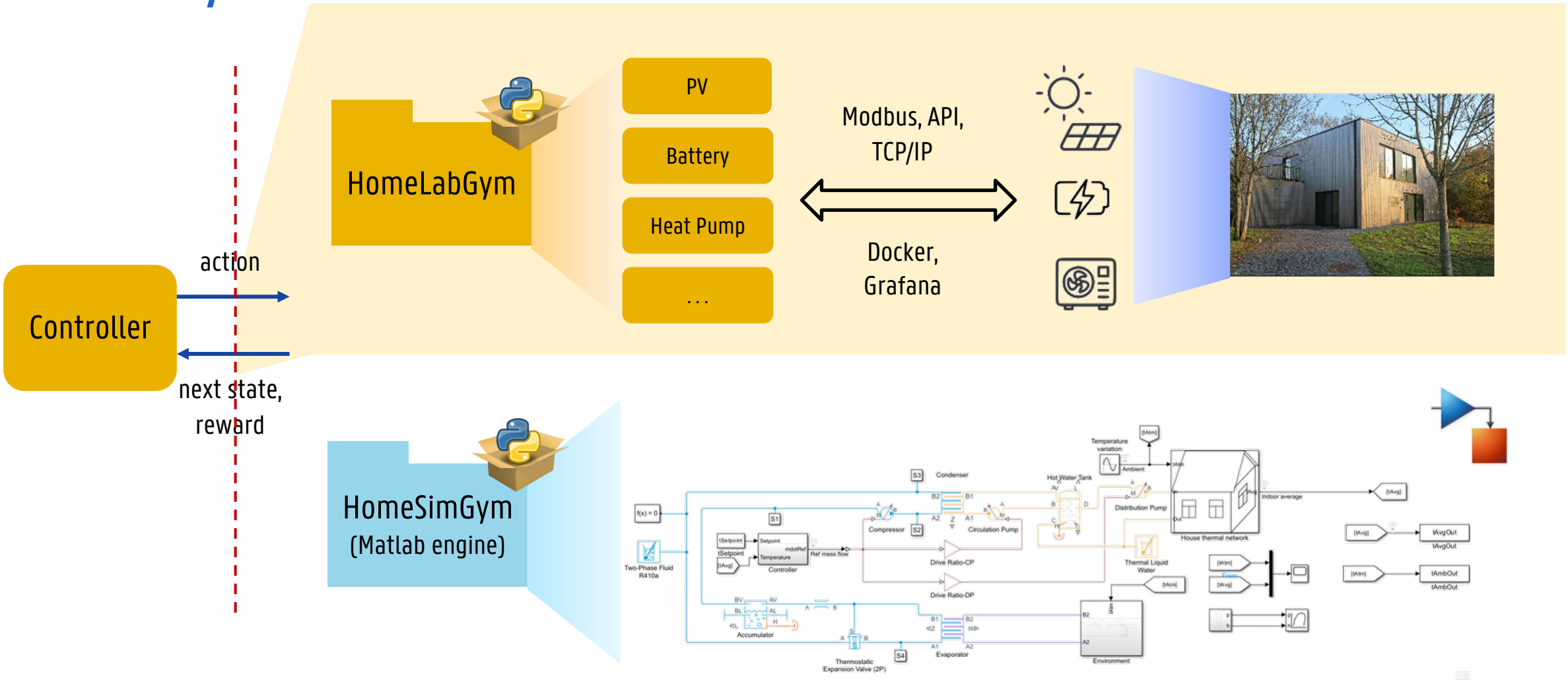
- Standardized approach
- Plug-and-play
- Extensible solution
- Broad range of communication protocols
- . . .



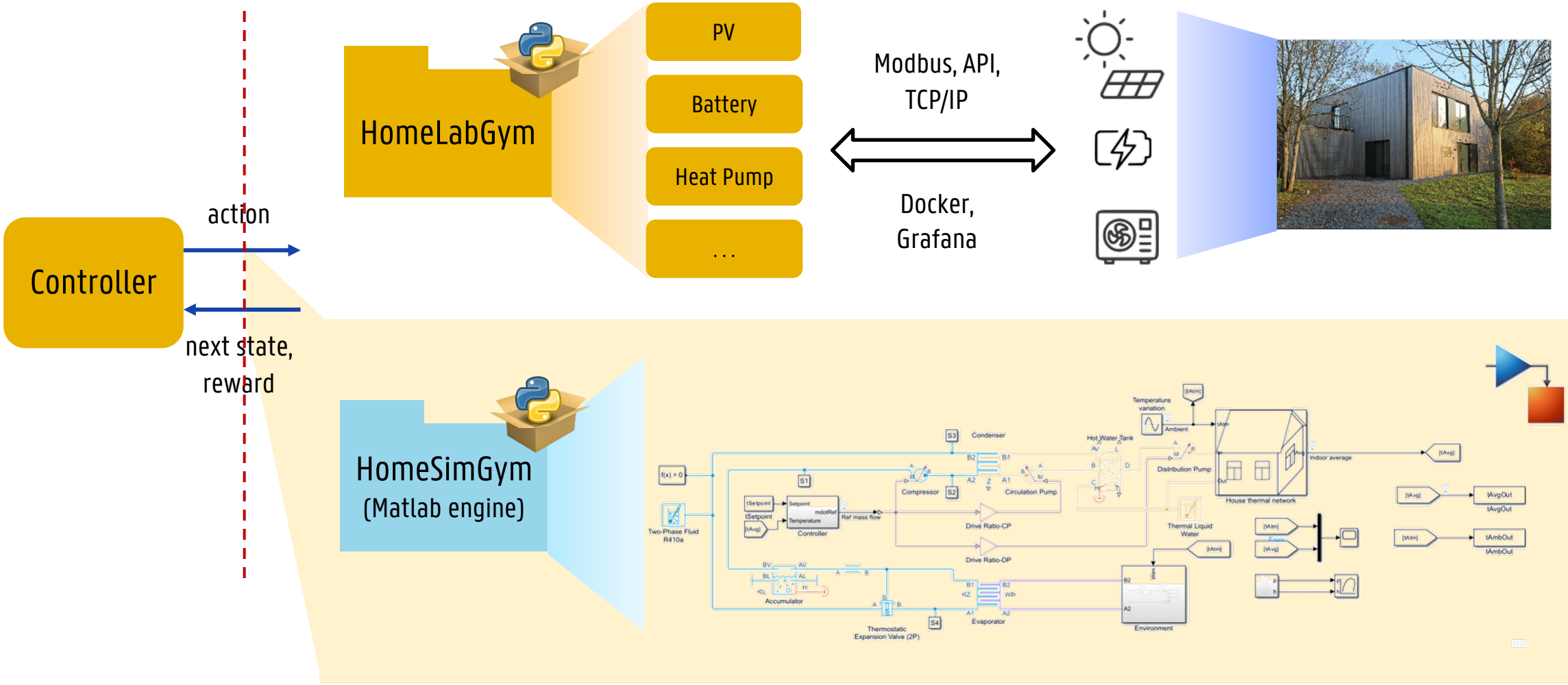
HomeLabGym


Controller
action
next state,
reward
HomeLabGym
PV
Battery
Heat Pump
...
Modbus, API,
TCP/IP
Docker,
Grafana
Gymnasium
Standardized interface, used for many RL benchmarks
(cf. OpenAI gym environments)

HomeLabGym


Controller
action
next state,
reward
HomeLabGym
PV
Battery
Heat Pump
...
Modbus, API,
TCP/IP
Docker,
Grafana
HomeSimGym
(Matlab engine)

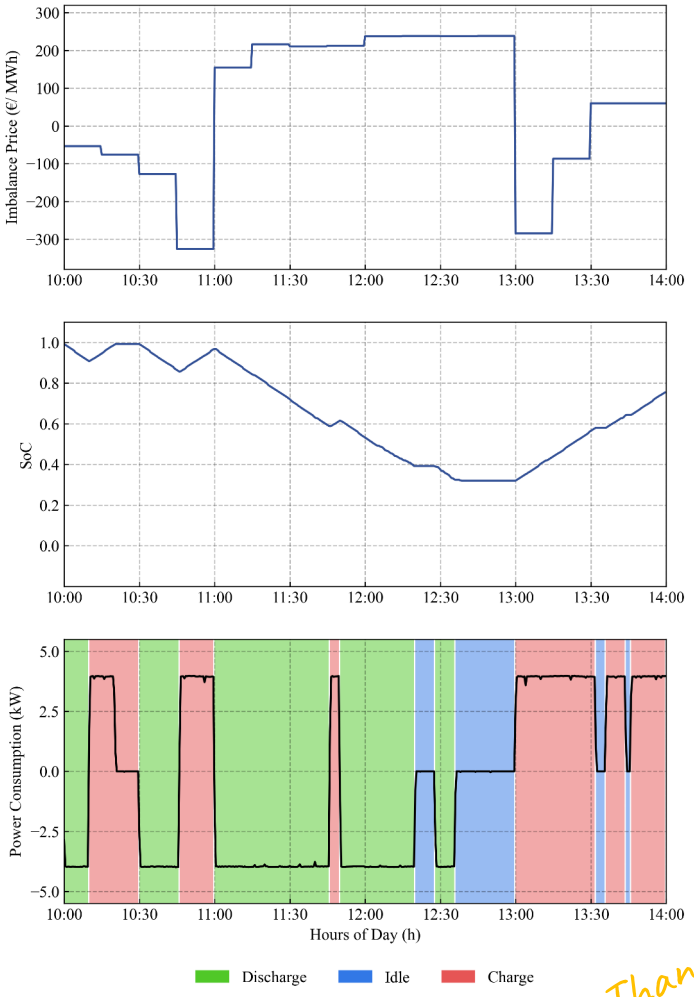
HomeLabGym


Controller
action
next state,
reward
HomeLabGym
PV
Battery
Heat Pump
...
Modbus, API,
TCP/IP
Docker,
Grafana
HomeSimGym
(Matlab engine)

Example use case for HomeLabGym: Validation of RL for energy arbitrage

S.S. Karimi Madahi, G. Gokhale, M.-S. Verwee, B. Claessens and C. Develder, **"Control policy correction framework for reinforcement learning-based energy arbitrage strategies"**, in Proc. 15th ACM Int. Conf. Future Energy Sys. (e-Energy 2024), Singapore, 4-7 Jun. 2024, pp. 1-9.. doi:10.1145/3632775.3661948

- Case study:
  - Home battery
  - Time-varying prices (qh-based)
  - Distributional DQN strategy accounting
- Validation of simulation
  - Real-world battery
  - Using HomeLabGym
- Results
  - **Latency** in communication (up to ~10s)
  - **Battery efficiency** affected by temperature
  - Effect on **daily revenue**: real profit €5.6 vs. simulated €6



Thanks Soroush!



Thank you!
Any questions?

chris.develder@ugent.be

http://users.ugent.be/~cdvelder

https://ugentai4e.github.io



BLUE BIRD

Funded by the European Union

The AI4E team at IDLab, Ghent University – imec



https://ugentai4e.github.io/